



FINAL EXAM PAPER
Quarter: 1 - Academic Year: 2025 – 2026
School of Computing and Information Technology

Course Code: CSW 439
Course Name: Data Mining
Type of Exam: Computer-Based Practical Exam

Exam Code: 01
Duration: 90 minutes
Maximum Mark: 100

Instructions:

- This paper consists of 3 questions. This paper carries 100 marks. Question 1 carries 35 marks, Question 2 carries 40 marks and question 3 carries 25 marks.
- Students are allowed to use their pre-prepared materials, including lecture notes, textbooks, quizzes, and lab solutions, which are saved in one compressed folder (.zip or .rar, etc.) prepared before the exam begins.
- Students are NOT allowed to use the Internet and any search or AI tools during the exam. Collaboration is NOT allowed.
- Bad coding style: 10 marks will be deducted from the exam.

Students are required to apply unsupervised learning techniques (k -means clustering) using Python to perform classification to predict the diabetes disease (diabetic, non-diabetic or predict-diabetic) based on medical information.

The data used is the file 'Dataset of Diabetes.csv'. The data consist of medical information, laboratory analysis. The data consist of medical information, laboratory analysis... etc. The data are: ID, No. of Patient, Age, Gender, Creatinine ratio(Cr), Body Mass Index (BMI), Urea, Cholesterol (Chol), LDL, VLDL, Triglycerides(TG) and HDL Cholesterol, HbA1C, Class (the patient's diabetes disease class may be Diabetic (Y), Non-Diabetic (N), or Predict-Diabetic (P)).

Question 1 (35 marks)

- a. Data Preprocessing (35 marks)

Question 2 (40 marks)

- a. Perform k -means clustering ($k = 3$) on the 80% training set to generate clusters (20% of data used for testing). (20 marks)
- b. Visualize the clusters (PCA-reduced features) (10 marks)
- c. Calculate and show the average Silhouette score (10 marks)

Question 3 (25 marks)

- a. Perform cluster - class label mapping: For each cluster, identify the most frequent ground truth class label among its members. (15 marks)
- b. Calculate and show the accuracy of the classification model using the test data. (5 marks)
- c. Repeat Question 2.a, 2.c with different values of k ($k = 4, 5, \dots$) and using Silhouette scores to determine the best value of k , named k^* , and show the accuracy of the classification model in the case $k = k^*$. (5 marks)

----- THE END -----

Lecturer's Signature

Phạm Đại Xuân